

CO2xN: Porewater SO4/Cl

July 2026 Samples

2026-09-29

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-09-23" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "July"
Site = "GCRew"
Project_name = "CO2xN"
run_notes = "Had to split 2M July into two runs, computer restarted and halted run.
Needed to re-run standard curve" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/CO2xN_202607_C1_1.txt"
raw_file_name_so4 = "Raw Data/CO2xN_202607_S04_1.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xN_PW_Processed_C1_S04_202607_1.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = here("Porewater QAQC Logs", "GCRew_S04_C1_QAQC_Log.csv")
```

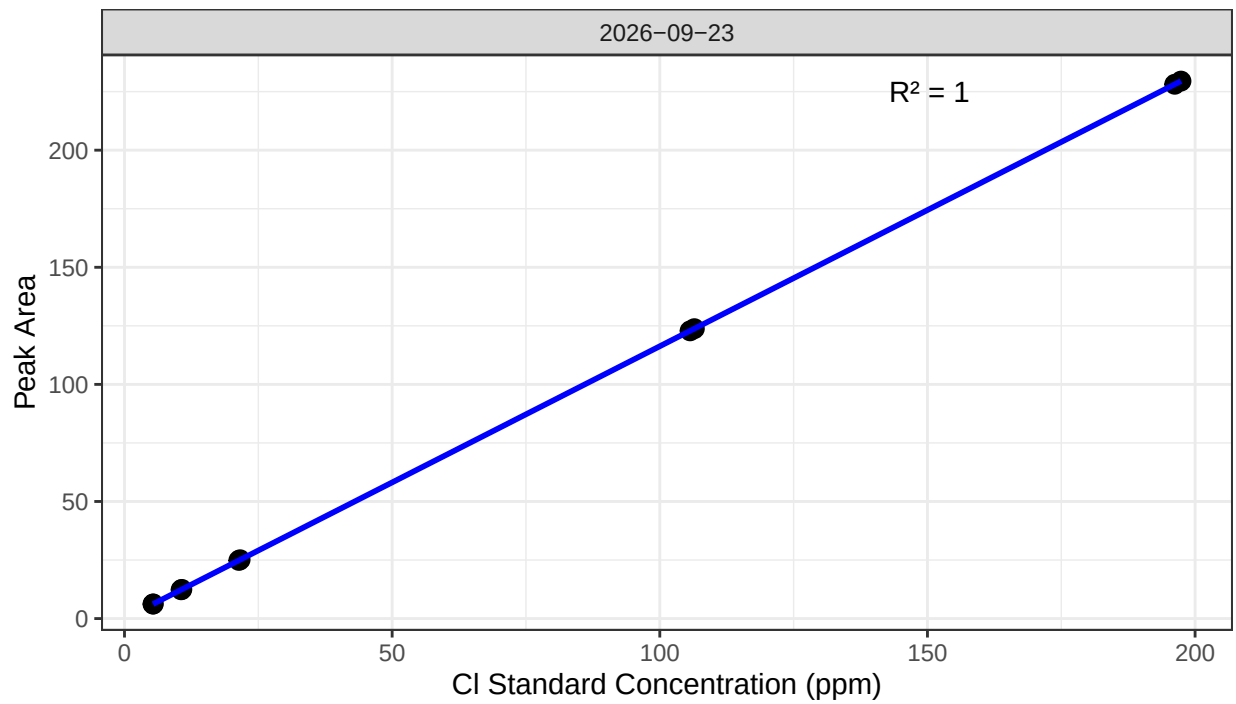
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

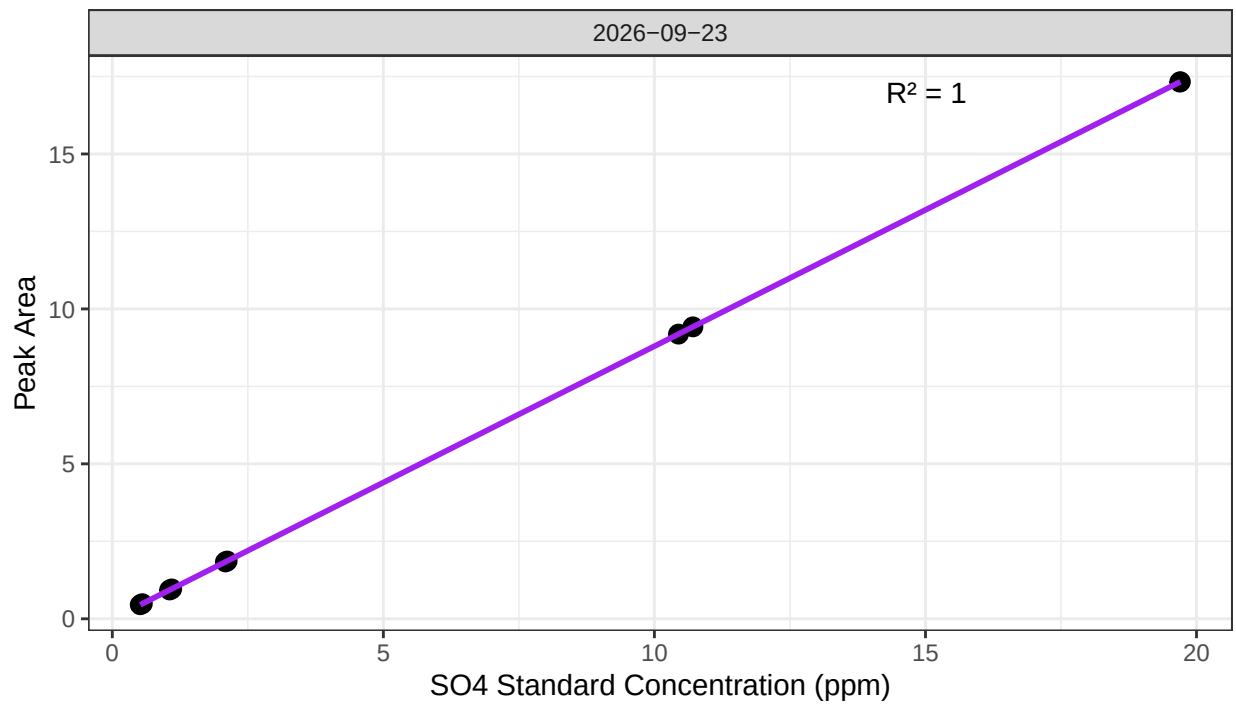
##Import Sample Data

0.2 Assess Standard Curves

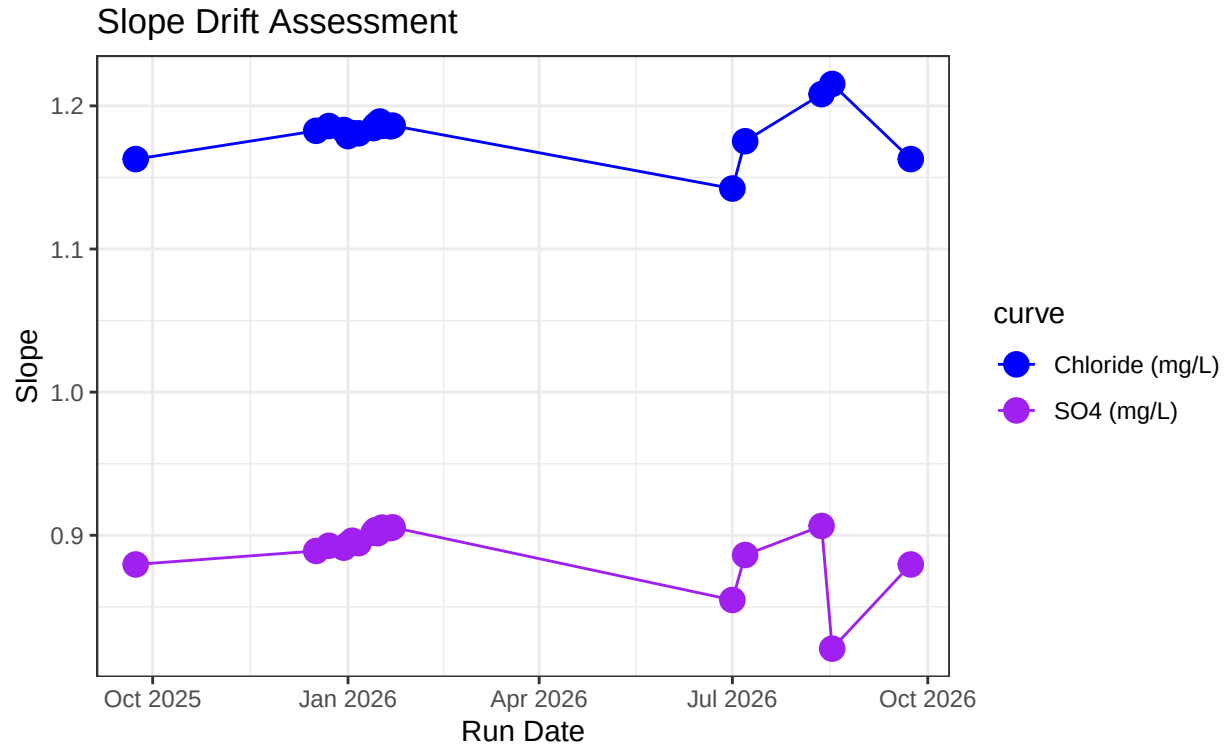
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

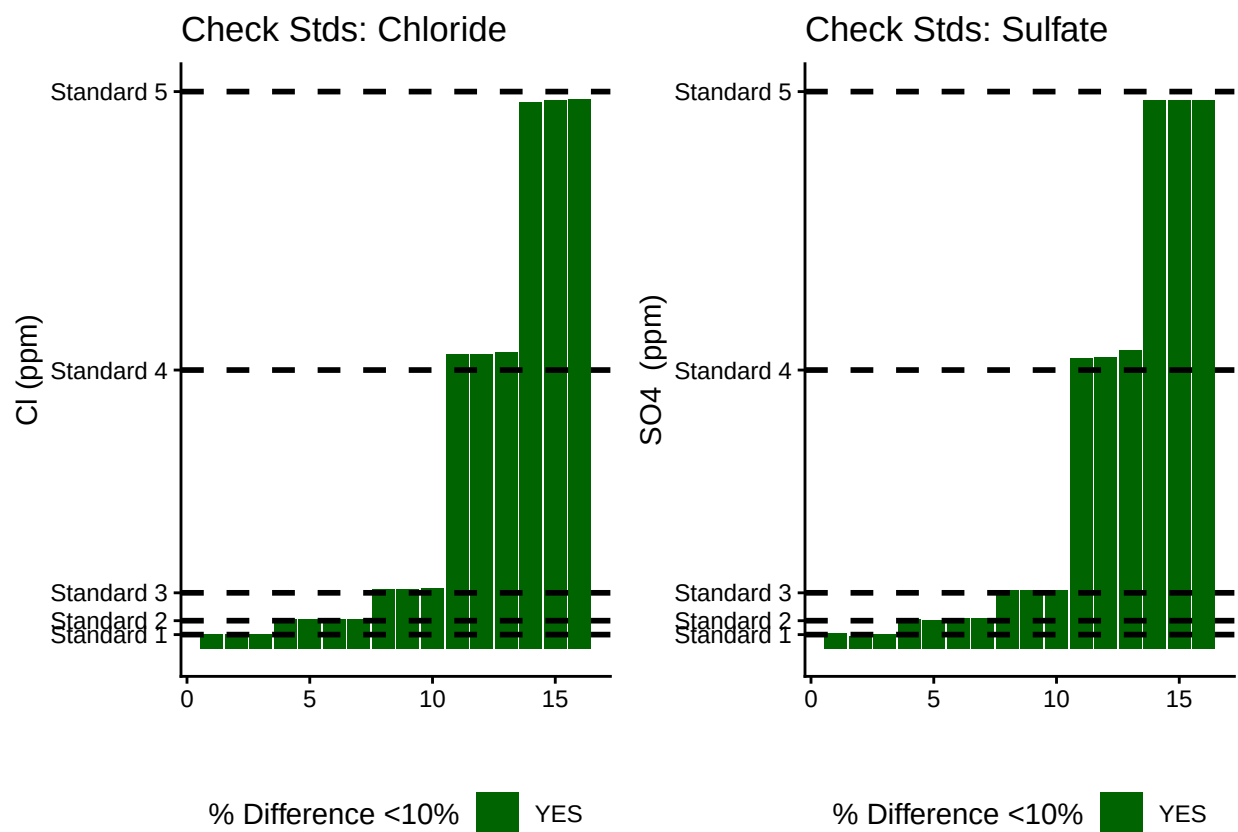
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 5.35 0.0232 0.00434 Chloride Check Standard RSD within Range - ~
## 2 Standard 2 10.7 0.0749 0.00702 Chloride Check Standard RSD within Range - ~
## 3 Standard 3 21.4 0.135 0.00631 Chloride Check Standard RSD within Range - ~
## 4 Standard 4 106. 0.419 0.00396 Chloride Check Standard RSD within Range - ~
## 5 Standard 5 197. 0.610 0.00310 Chloride Check Standard RSD within Range - ~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.504 0.0565 0.112 Sulfate Check Standard RSD within Range --
## 2 Standard 2 1.07 0.0333 0.0312 Sulfate Check Standard RSD within Range --
## 3 Standard 3 2.10 0.0182 0.00870 Sulfate Check Standard RSD within Range --
## 4 Standard 4 10.5 0.148 0.0140 Sulfate Check Standard RSD within Range --
## 5 Standard 5 19.7 0.0109 0.000551 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



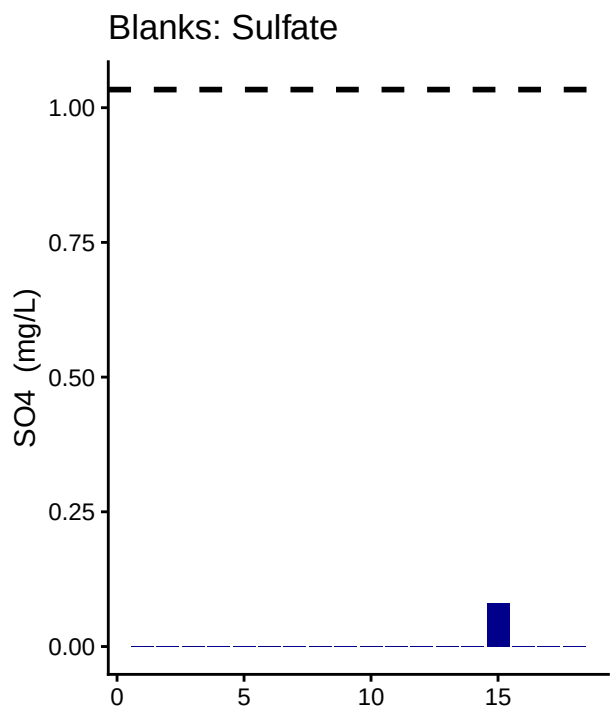
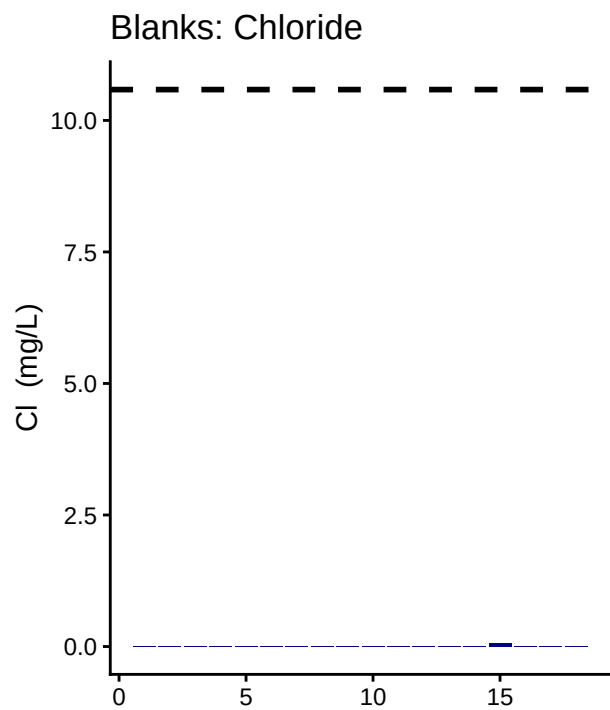
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ Y

Blank Conc <25% Quartile Samples ■ Y

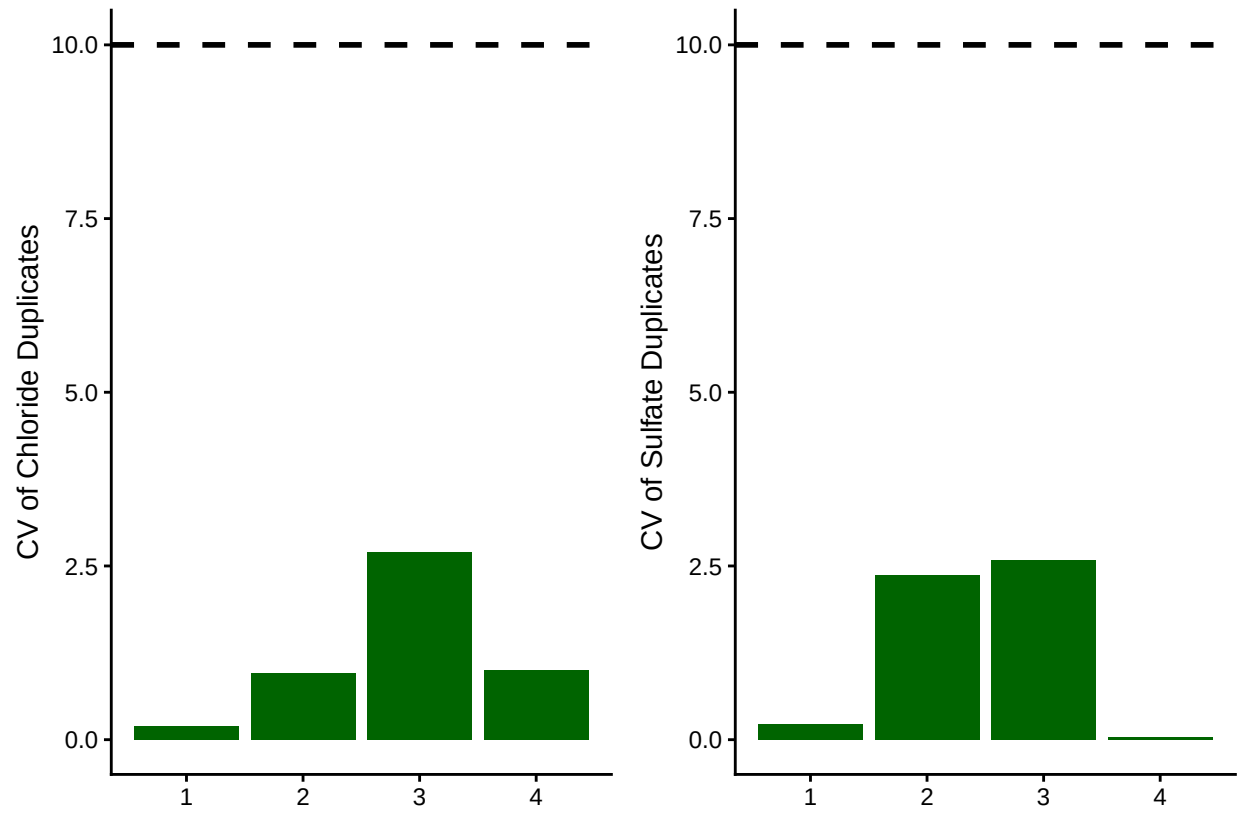
```
## Chloride blanks mean ppm:
```

```
## [1] 0.005322222
```

```
## Sulfate blanks mean ppm:
```

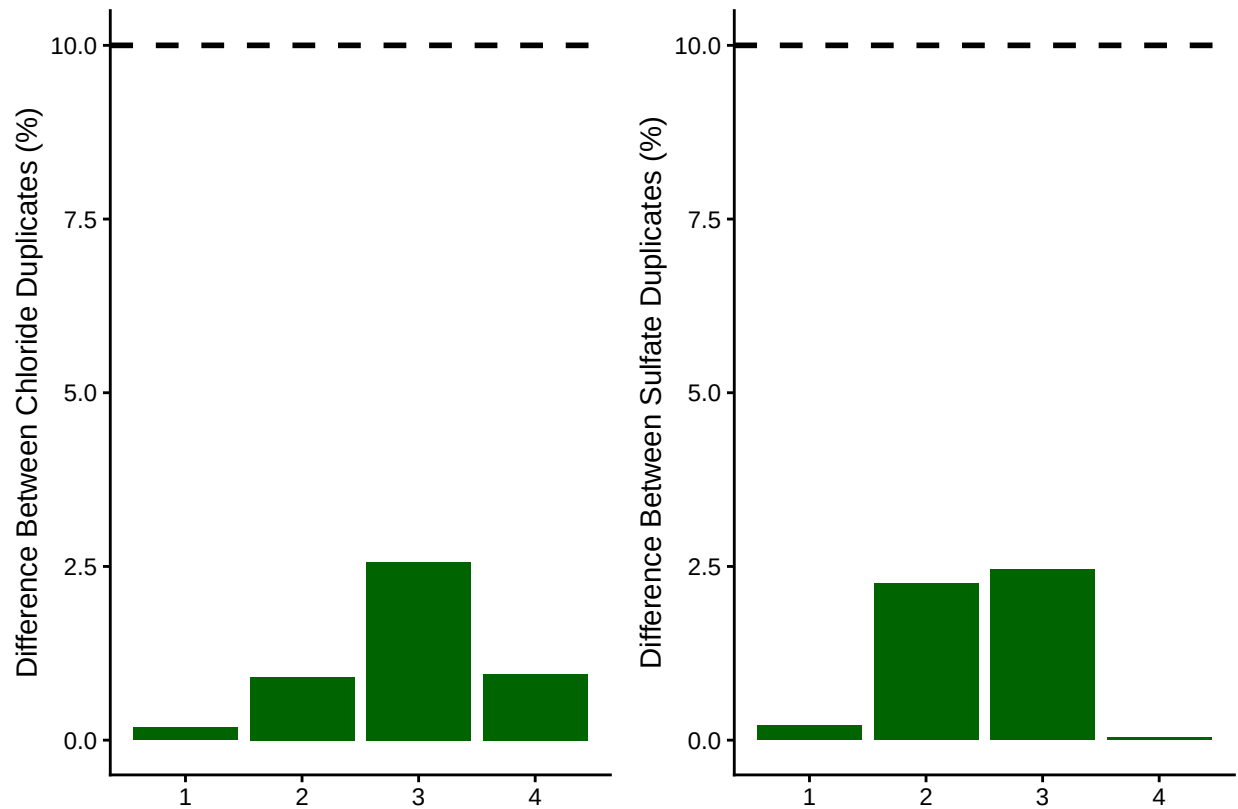
```
## [1] 0.004566667
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

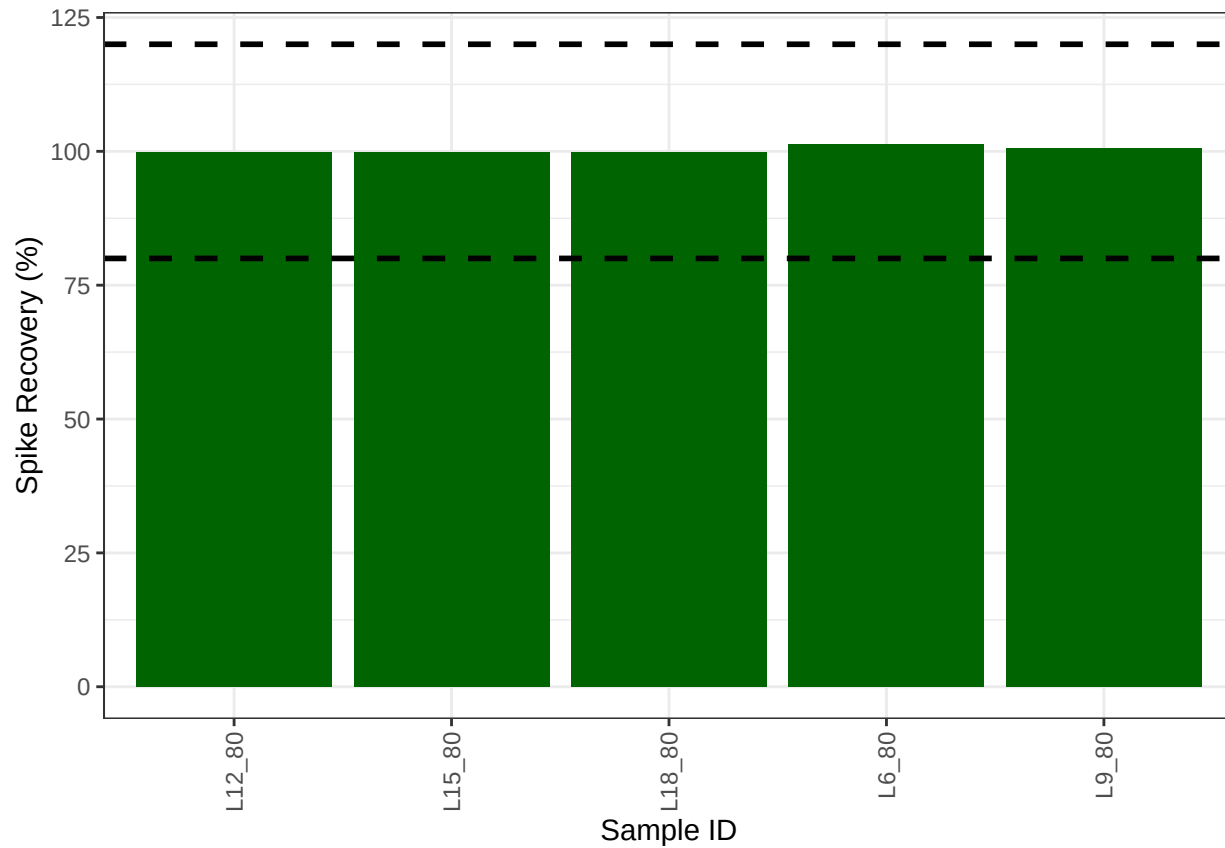
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807 * Cl_ppm) + 0.026) / 1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

# Need to determine dilution factors for your samples
# for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

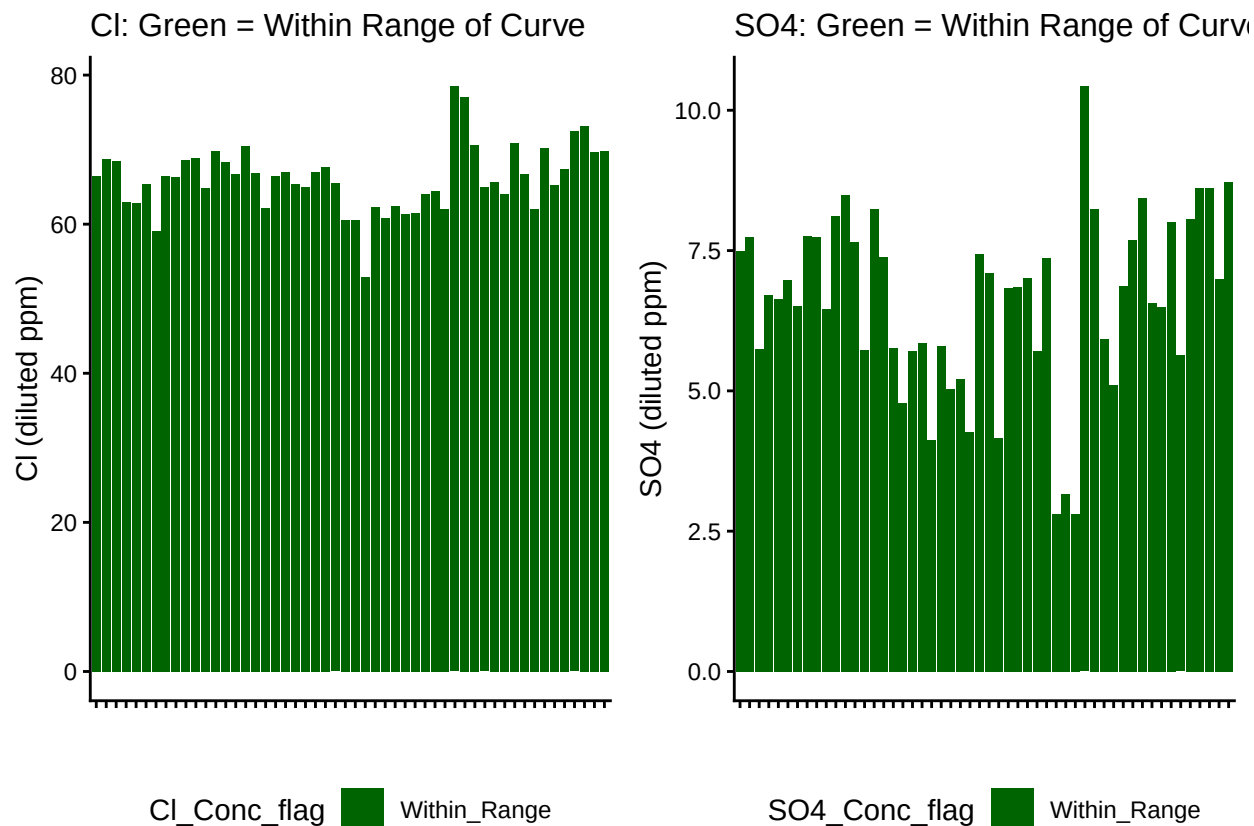


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

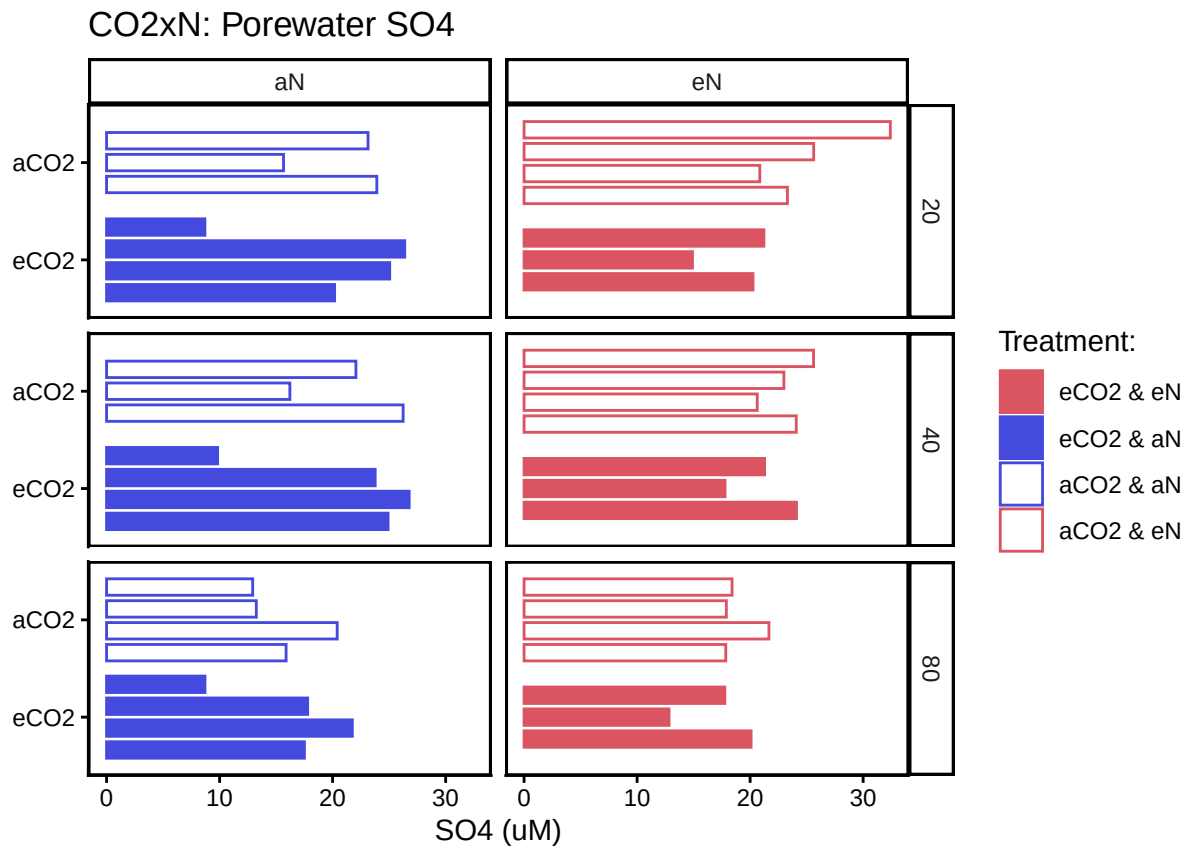
Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Check to see if samples run match metadata & merge info

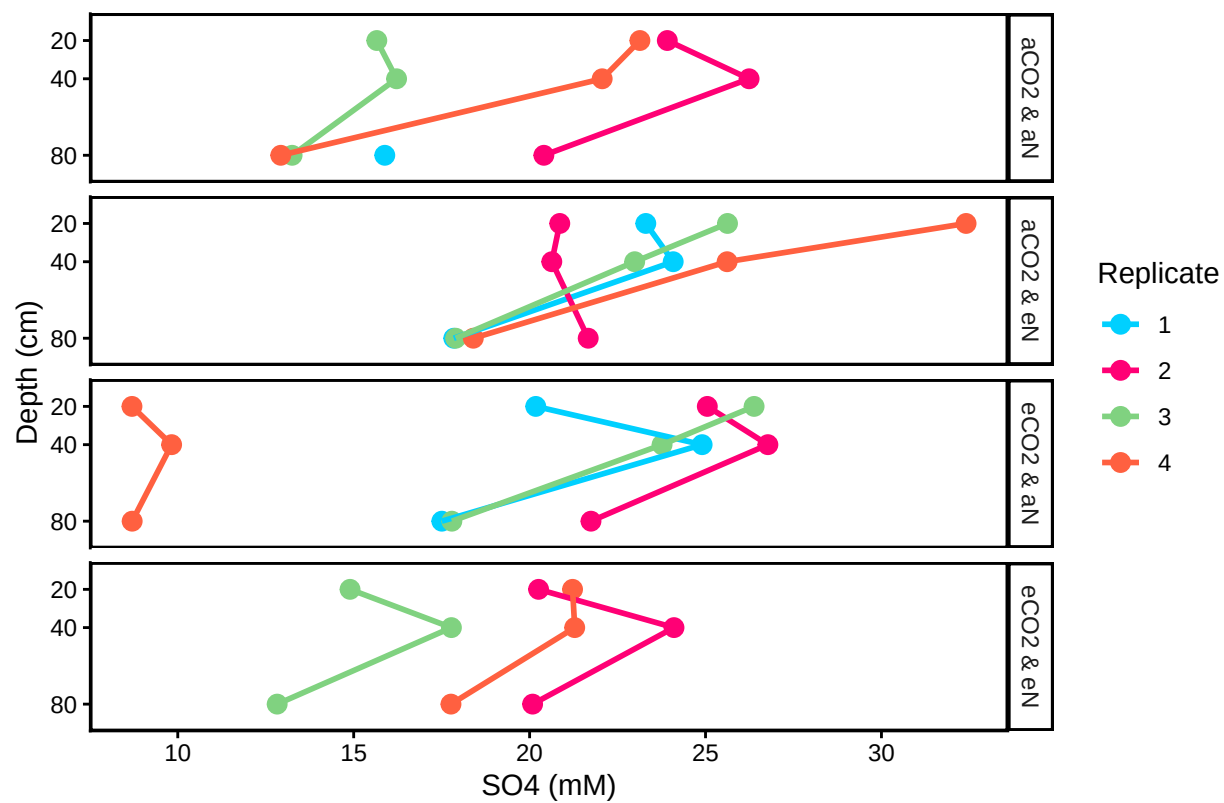
```
## All sample IDs are present in metadata.
```

0.10 Organize Data

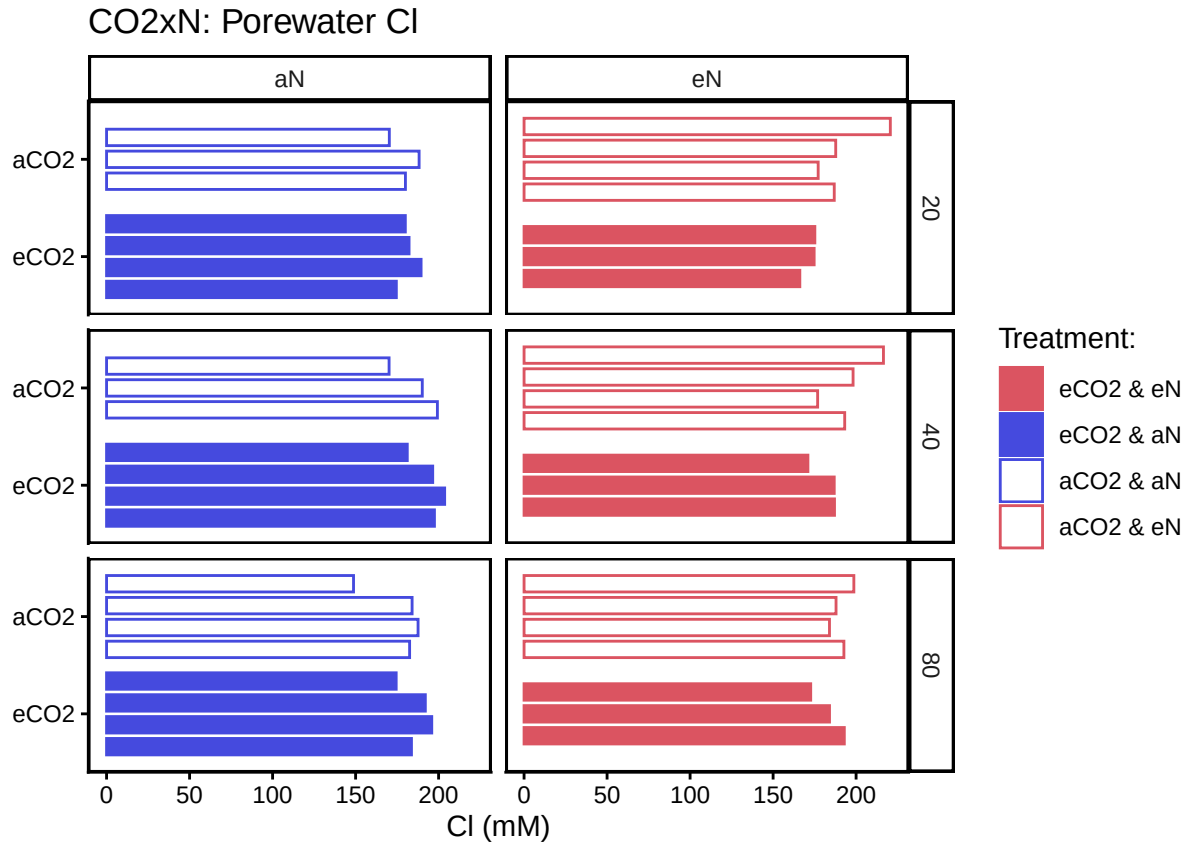
0.11 Visualize Data SO4

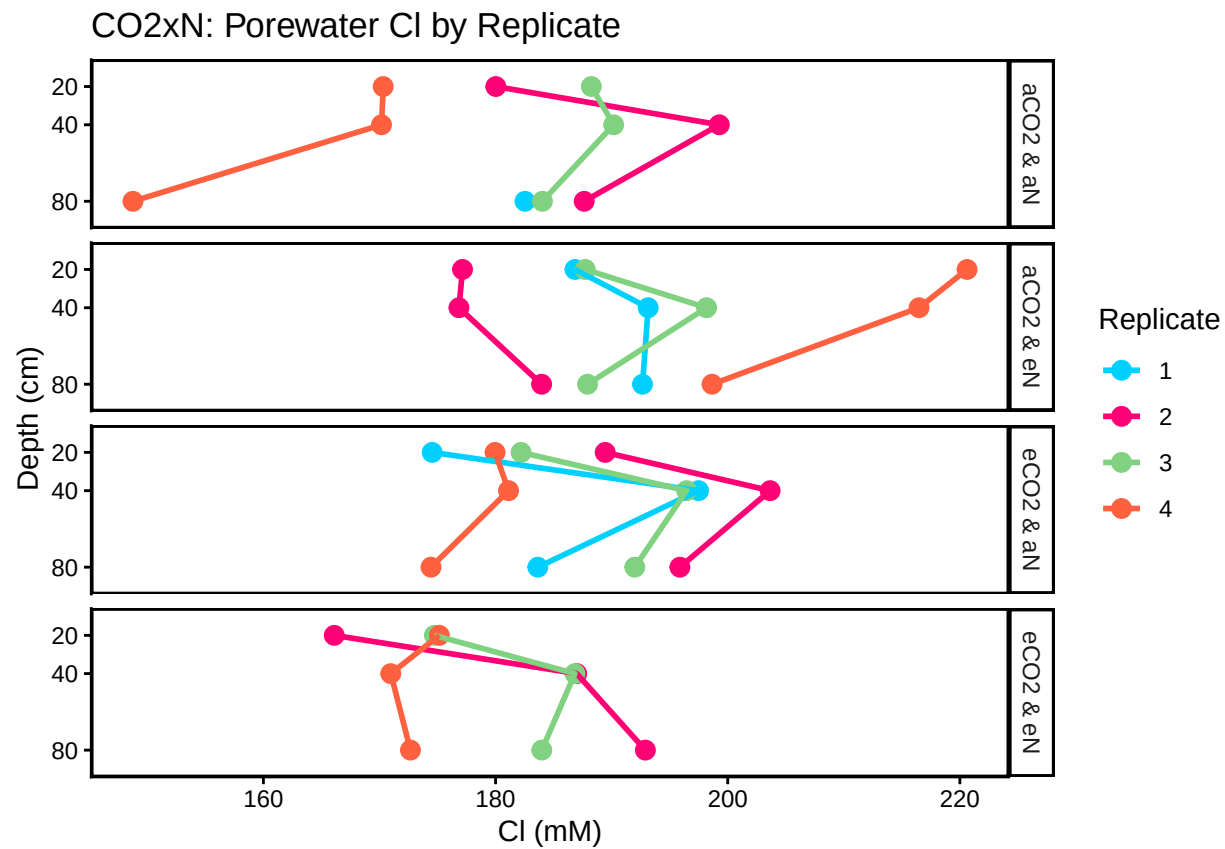


CO2xN: Porewater SO4 by Replicate



0.12 Visualize Data CI





0.13 Export Processed Data

#end